Tyler N. **Morrison**

Physical AI Research Engineer

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Experience

Boston Dynamics
Boston, MA

Staff Research Engineer - Reinforcement Learning
February 2026 — Present

- Trained and deployed **hierarchical RL** models for complex robotic manipulation tasks
- **Designed ML model export pipeline** for edge-device inference
- Contributed to team **research investigations in RL for dextrous manipulation**

Agility Robotics
Pittsburgh, PA and Remote

Senior Robotics Software Engineer
February 2022 — January 2026

- **Identified and solved novel technical problems in robotics, drove product-market fit, and scaled engineering for first five customer use cases.**
- **Led peers in a highly dynamic fast-paced start-up environment including company growth through multiple funding rounds.**
- Named **technical lead** of the Skill Development team of six which uses primitives to construct robot behaviors for use by Applications.
- **Technical lead** of a year-long critical inter-service communication project spanning from cloud to robot.
- Led two critical **world-first** commercial deployments (link) as **technical lead and interim manager**.
- **Technical lead and demo lead for over 140 robot-hours of continuous, live humanoid tradeshow demos** including the MODEX 2024 (link) and Promat 2025 live demos, and key individual contributor for Promat 2023.
- Engaged with technical leads from controls and AI groups to determine requirements for business-critical technology development projects.
- Led international **high-impact demos** for engagements with press, high net worth individuals, Fortune 100 executives, investors, etc., resulting in 100's of millions of USD in raised capital, two strategic partnerships, and two pilot projects.
- **Product development:** Drove after-action product gap analyses for engineering leadership after deployments and customer engagements to ensure product-market fit.
- Developed software bug triage process which reduced the backlog of open bugs by over 50% over two years.
- **Humanoid robot behavior developer** for a dozen unique use-cases.
- Constructed our first whole-robot hardware-in-the-loop test pipeline for continuous integration.
- ***Cross-functional developer:*** Primary contributor for the cloud-to-robot interface for managing robot behavior commands. Key contributor to architecture definition phase of forthcoming generational hardware design revision.
- Architect and lead developer for CLI tool for joint workflow/behavior development and debugging in use by 20+ developers for years
- Integrated **object detection**, **manipulation**, and **navigation** systems into complex tote manipulation behaviors deployed with >99% task success and 0 major faults in 40 hours of runtime across ten robots.
- ***T-shaped engineer:*** Developed and shipped features for robotics software in areas throughout the stack including: **perception**, **manipulation**, **logging and diagnostics**, **cloud connectivity**, **hardware-in-the-loop testing**, and more...

Design, Innovation and Simulation Lab (DISL)
The Ohio State University

Graduate Research Assistant
August 2017 — January 2022

- Developed simulation tools for powered ankle prosthesis design using trajectory optimization for human gait adaptation prediction.
- Modeled, tested, and analyzed variable stiffness robotic arm links for use in physical human-robot interaction.
- Mentored high school, undergraduate, and MS student research projects.

NSF Interfaces and Surfaces REU
Clemson University

REU Research Assistant
May 2016 — August 2016

Education

The Ohio State University
Columbus, OH, USA

Ph.D. Candidate in Mechanical Engineering
Aug. 2017 — Dec. 2022

- Dissertation: *Computational Design Methods for Compliant Robotic Ankle Prostheses*
- Distinguished University Fellow & Dept. Award Recipient

The University of Tulsa
Tulsa, OK, USA

B.S. Mechanical Engineering
Aug. 2013 — May 2017

- Chapman Presidential Scholar

Skills and Previous Experience

Software Bazel, Python, JavaScript, C/C++, Docker, Bash, Rust, HTTP, AWS, MQTT
Robotics Behavior Trees, Object Detection†, Factor Graphs†, Kalman-filters†, IK, RL control†, Teleop, Demonstration learning
Modeling & Design Blender, Solidworks, AutoCAD, Figma, Adobe Illustrator
Simulation Mujoco, OpenSim, ANSYS, ABAQUS
Hardware Arduino, Raspberry Pi, DC Motors, Lidar, Binocular depth†, Basic Circuit Design, EtherCAT†
Soft Technical leadership, Hiring, Interim management, High-pressure demos, Executive-level communication
Languages English, Modern Greek, Spanish

†familiar with as a consumer

Selected Academic Publications

Tyler Morrison, Hai-Jun Su. *Stiffness Modeling of a Variable Stiffness Compliant Link*. Mechanism and Machine Theory. 2020. «Published». DOI: 10.1016/j.mechmachtheory.2020.104021

Tyler Morrison, Richard Jagacinski, Jordan Petrov. *Drivers' Attention to Preview and Its Momentary Persistence*. IEEE Transactions on Human-Machine Systems. 2023. «Published». DOI: 10.1109/THMS.2023.3266152

Tyler Morrison, Chunhui Li, Xu Pei, Hai-Jun Su. *A Novel Rotating Beam Link for Variable Stiffness Robotics Arms*. IEEE International Conference on Robotics and Automation. *Montreal, CA*. May 2019. «Published». DOI: 10.1109/ICRA.2019.8793833

Tyler Morrison, Hai-Jun Su. *Gait Prediction for Prosthesis Design Evaluation*. ASME IDETC-CIE. *St. Louis, MO, USA*. November 2022. «Published». DOI: 10.1115/DETC2022-89739

Tyler Morrison, Dylan Trainor, Hai-Jun Su. *Optimization of the Compliant Drive Mechanism for a Prosthetic Ankle*. ASME IDETC-CIE. *St. Louis, MO, USA*. August 2020. «Published». DOI: 10.1115/DETC2020-22442

Selected Honors & Awards

Graduate

2017 **Distinguished University Fellow**, The Ohio State University *Columbus, OH, USA*
2017 **Department Supplementary Fellowship Award**, The Ohio State University *Columbus, OH, USA*

Undergraduate

2017 **College of Engineering and Natural Sciences Steven J. Bellovich Medal**, The University of Tulsa *Tulsa, OK, USA*
2017 **Senior Project – Most Valuable Team Member**, The University of Tulsa, Mech. Eng. Dept. *Tulsa, OK, USA*
2016 **Putnam Competition Team Member**, The University of Tulsa, Math Dept. *Tulsa, OK, USA*
2016 **Nominee, National Barry Goldwater Scholarship**, The University of Tulsa *Tulsa, OK, USA*
2013 **Presidential Scholar**, The University of Tulsa *Tulsa, OK, USA*

Other

2011 **Eagle Scout**, Boy Scouts of America *Kansas, USA*

Passions

- Long-distance running and cycle touring: Multiple centuries completed including 200 mile solo across Ohio.
- Dog training: Adopted and rehabilitating a beloved reactive dog.
- Language learning and cultural exchange: Individual study in modern Greek to advanced level.